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Project Presentation of Industry Oriented Hands on Experience (22CS422)
On

Social Media Analytics for Trend Prediction Using Machine Learning

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Introduction

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The Problem

500M+ social media posts generated daily

Traditional trend detection methods:
- Rely on periodic surveys
- Use retrospective historical data
- Cannot capture real-time dynamics
- Are slow and non-adaptive

Result: Missed opportunities for marketers, journalists & crisis communicators

Our Solution

ML-Powered Trend Prediction System

Integrates:
- NLP Preprocessing Pipeline
- TF-IDF Feature Extraction
- Hashtag Velocity Analysis
- Engagement-Based Metrics

Models Compared:
- Logistic Regression
- Naive Bayes
- Random Forest (Best: 91%)

Literature Review

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Frequency Thresholding

Early trend detection relied on burst detection and keyword frequency analysis (Kleinberg, 2002).

TwitterMonitor

Clustered bursty keywords into real-time topics using streaming algorithms (Mathioudakis & Koudas, 2010).

LDA Topic Modelling

Unsupervised topic discovery but does not optimise for prediction (Blei et al., 2003).

Sentiment Analysis

TF-IDF + supervised ML entered trend analytics via sentiment classification (Pang & Lee, 2008).

Engagement Signals

Like/retweet counts correlate strongly with virality and trending thresholds (Hong & Davison, 2010).

BERT / Transformers

State-of-the-art contextual embeddings at significantly higher computational cost (Devlin et al., 2019).

Methodology & Dataset

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Data Collection → Pre-processing → Feature Engineering → Model Training → Evaluation

Dataset

Source: Twitter Academic Research API

Size: ~150,000 posts (6 months)

Domains: Technology, Entertainment, Politics, Sports, Public Health

Features: Timestamp, text, hashtags, likes, retweets, trending label

Class Split: 31% trending | 69% non-trending

Preprocessing Pipeline

1. URL & mention removal
2. Lowercase normalisation
3. Punctuation stripping
4. Whitespace tokenisation
5. NLTK stop-word removal
6. WordNet lemmatisation

Feature Engineering

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TF-IDF Textual Features

- Max 10,000 unigrams & bigrams
- Amplifies domain-specific discriminative vocabulary
- Weighted by inverse document frequency

Hashtag Velocity

- Unique hashtag count per post
- 24-hour usage frequency tracking
- Frequency rate-of-change (velocity spike = imminent trend)

Engagement Metrics

- Like count (min-max normalised)
- Retweet count (min-max normalised)
- Retweet-to-like ratio

Tools & Technologies

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Language	Python 3.10
Data	Pandas, NumPy
Visualisation	Matplotlib, Seaborn
IDE	Jupyter Notebook, VS Code
ML Library	Scikit-learn 1.3
NLP	NLTK, VADER
API	Twitter API + Tweepy
Version Control	Git / GitHub

Results & Analysis

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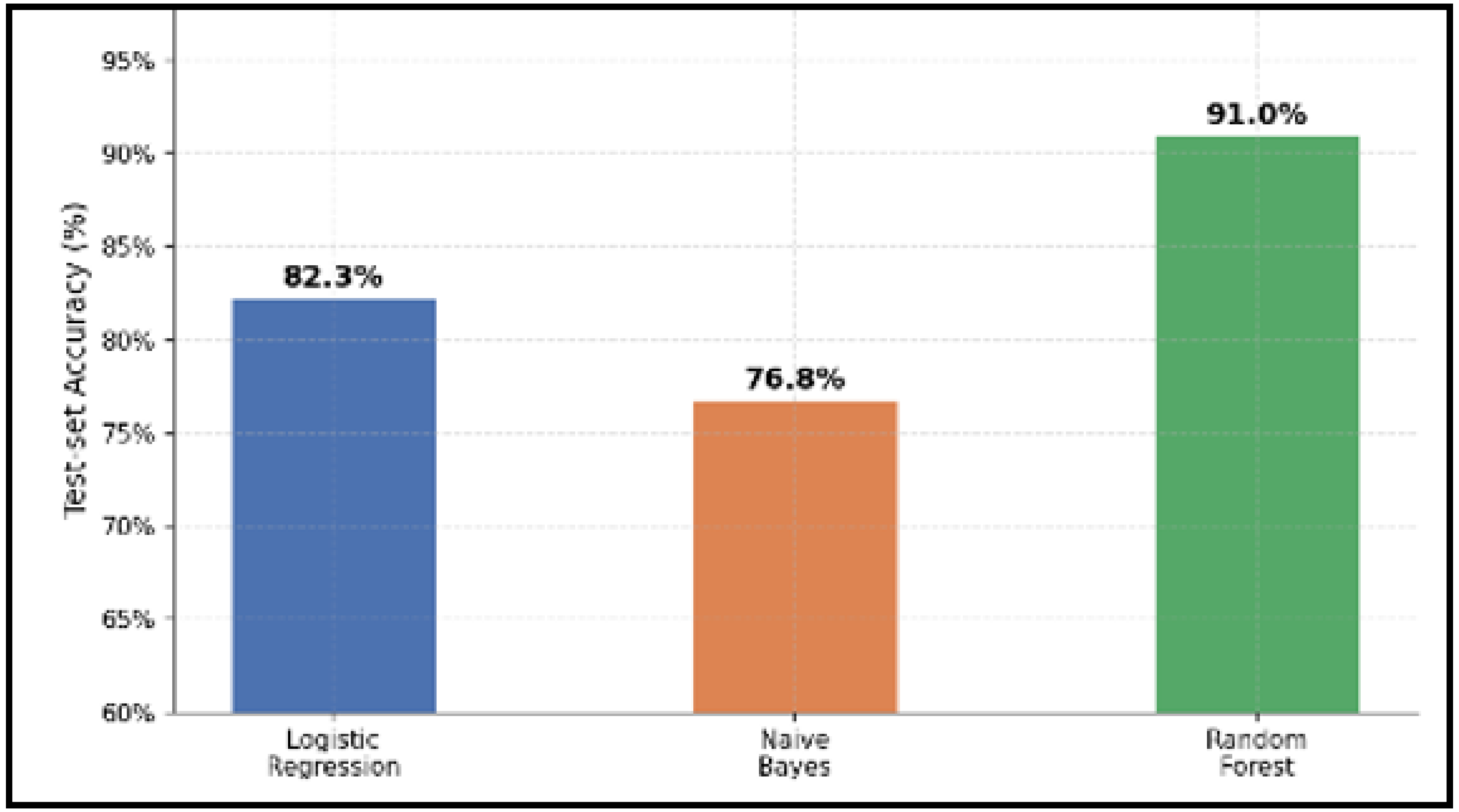

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
Logistic Reg.	82.3	80.1	78.6	79.3
Naive Bayes	76.8	74.5	73.2	73.8
Random Forest	91.0	89.5	88.7	89.1

91%

Best Accuracy (Random Forest)

4.2h

Early Warning Before Official Trend

±0.55 %

CV Variance (5-Fold)

Conclusion & Future Scope

Conclusion

- Random Forest achieves 91.0% test accuracy — best among all three models
- System provides ~4.2-hour early warning before official trend onset
- Hashtag velocity, retweet count & sentiment polarity are the dominant predictors
- Demonstrates clear practical value for digital marketing, PR, and newsrooms

Future Scope

- BERTweet / RoBERTa for contextual text features
- Kafka + Spark Streaming for real-time deployment
- Multilingual & cross-platform extension (Instagram, Reddit)
- Graph-based retweet-propagation features
- Adversarial bot-filtering techniques

Thank You